Prayas JEE AIR (2025)

WEEKLY TEST 31

DURATION : 180 Minutes | **DATE : 16/01/2025** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (1)
3. (4)
4. (1)
5. (1)
6. (4)
7. (2)
8. (4)
9. (4)
10. (2)
11. (4)
12. (2)
13. (3)
14. (3)
15. (2)
16. (4)
17. (4)
18. (4)
19. (1)
20. (1)
21. (2)
22. (6)
23. (3)
24. (4)
25. (25)

CHEMISTRY

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28. (1)
29. (1)
30. (1)
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32. (3)
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35. (2)
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40. (3)
41 (2)
42 (1)
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44. (1)
45. (2)
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MATHEMATICS

51. (4)
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60. (2)
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70. (3)
71. (4)
72. (8)
73. (3)
74. (5)
75. (3)

SECTION-I (PHYSICS)

1. **(2)**

$$\frac{1}{2}gt^2 = h \Rightarrow t = \sqrt{\frac{2h}{g}}$$

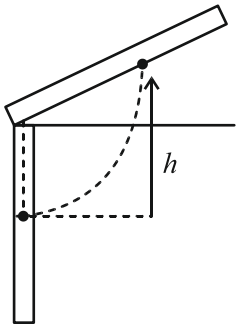
$$v_0 - a_0 t = 0 \Rightarrow \frac{v_0^2}{a_0^2} = \frac{2h}{g}$$

$$\Rightarrow h = \frac{v_0^2 g}{2a_0^2}$$

2. **(1)**

1st reaction is fusion and 4th reaction is fission.

3. **(4)**



$$T.E_i = T.E_f$$

$$\frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2} \times \frac{1}{3}m\ell^2\omega^2 = mgh$$

$$\Rightarrow h = \frac{1}{6}\frac{\ell^2\omega^2}{g}$$

4. **(1)**

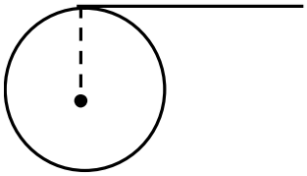
$$x_{cm} = \frac{\int dmx}{\int dm} = \frac{\int \lambda dx \cdot x}{\int dm}$$

$$= \frac{\int k\left(\frac{x}{L}\right)^n \cdot xdx}{\int k\left(\frac{x}{L}\right)^n dx} = \left[\frac{\frac{kx^{n+2}}{(n+2)L^n}}{\frac{kx^{n+1}}{(n+1)L^n}}\right]_0^L$$

$$= \left[\frac{x(n+1)}{n+2}\right]_0^L$$

$$x_{cm} = \frac{L}{2}, \frac{2L}{3}, \frac{3L}{4}, \frac{4L}{5}, \frac{5L}{6}, \ldots$$

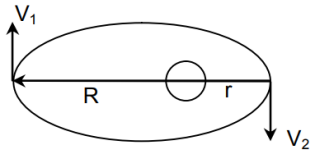
5. **(1)**



$$I = mr^2 + \frac{m(2r)^2}{12} + m(\sqrt{2r})^2$$

$$\Rightarrow I = 3mr^2 + \frac{mr^2}{3} = \frac{10mr^2}{3}$$

6. **(4)**



By conservation of angular momentum,

$mv_1R = mv_2r$

$\Rightarrow \quad v_1R = v_2r$

By conservation of energy,

$$-\frac{GMm}{R} + \frac{1}{2}mV_1^2 = -\frac{GMm}{r} + \frac{1}{2}mV_2^2$$

$$\Rightarrow V_1 = \sqrt{\frac{2GMr}{R(R+r)}}$$

Now, $mv_1R = m\sqrt{\frac{2GMrR}{r+R}}$

7. **(2)**

$\rho_{oil} < \rho < \rho_{water}$

Oil is the least dense of them so it should settle at the top with water at the base. Now the ball is denser than oil but less denser than water. So, it will sink through oil but will not sink in water. So it will stay at the oil-water interface.

8. **(4)**

$$A_1\ell_1 = A_2\ell_2 \Rightarrow \ell_2 = \frac{A_1\ell_1}{A_2} = \frac{A \times \ell_1}{3A} = \frac{\ell_1}{3} \Rightarrow \frac{\ell_1}{\ell_2} = 3$$

$$\Delta x_1 = \frac{F_1}{A_\gamma} \times \ell_1$$

$$\Delta x_2 = \frac{F_2}{3A_\gamma}\ell_2$$

Here $\Delta x_1 = \Delta x_2$

$$\frac{F_2}{3A\gamma}\ell_2 = \frac{F_1}{A\gamma}\ell_1$$

$$F_2 = 3F_1 \times \frac{\ell_1}{\ell_2} = 3F_1 \times 3 = 9F$$

9. **(4)**

$\Rightarrow (230 + 20) \times 1 \times 5 = m_{eq} \times 1 \times 50$

$m_{eq} = 25$ gm

10. **(2)**

Thermal energy corresponds to internal energy

Mass = 1 kg

Density = 4 kg/m^3

$\Rightarrow$ Volume $= \dfrac{\text{mass}}{\text{density}} = \dfrac{1}{4}m^3$

Pressure = 8×10^4 N/m^2

$\therefore$ Internal Energy $= \dfrac{5}{2}P \times V = 5 \times 10^4$ J

11. (4)

$\Rightarrow ma = -(kx + 2kx' \cos 60°)$

$\Rightarrow x' = x/2$

$\Rightarrow ma = -\dfrac{3kx}{2}$

$\Rightarrow a = -\left(\dfrac{3k}{2m}\right)x$

$T = 2\pi\sqrt{\dfrac{2m}{3k}}$

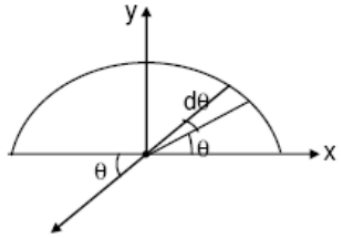
12. (2)

$V = n\lambda$

$\sqrt{\dfrac{T}{M}} = n(2\Delta x) \Rightarrow \sqrt{T} \propto \Delta x$

If tension is twice then Δx will increase to $\sqrt{2}\,\Delta x$.

13. (3)



Linear charge density $\lambda = \left(\dfrac{q}{\pi r}\right)$

$E = \int dE \sin\theta(-\hat{j}) = \int \dfrac{K \cdot dq}{r^2}\sin\theta(-\hat{j})$

$E = \dfrac{K}{r^2}\int \dfrac{qr}{\pi r}d\theta \sin\theta(-\hat{j})$

$= \dfrac{K}{r^2}\dfrac{q}{\pi}\int_0^{\pi}\sin\theta\, d\theta(-\hat{j}) = \dfrac{q}{2\pi^2\varepsilon_0 r^2}(-\hat{j})$

14. (3)

Apply gaus theorem

$E(4\pi r^2) = \dfrac{q_{in}}{\epsilon_0}$

$\Rightarrow E = \dfrac{q}{4\pi\epsilon_0 r^2} \Rightarrow E \propto \dfrac{1}{r^2}$

$\Rightarrow$ field depends only on r.

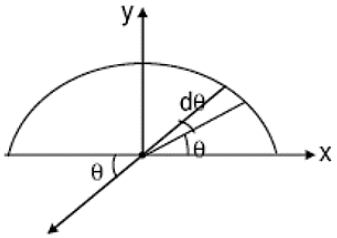
15. (2)

$V_c = E(1 - e^{-1/Rc})$

$1 - e^{-t/Rc} = \dfrac{120}{200} = \dfrac{3}{5}$

$\Rightarrow R = \dfrac{5}{1.84 \times 10^{-6}} = 2.7 \times 10^6 \Omega$

16. (4)



$d\vec{B} = \dfrac{\mu_0 di}{2\pi R}[-\cos\theta\hat{i} - \sin\theta\, j]$

$di = \dfrac{T}{\pi R}Rd\theta = \dfrac{1}{\pi}d\theta$

$d\vec{B} = \dfrac{\mu_0 I}{2\pi^2 R}[-\cos\theta\hat{i} - \sin\theta\, j]$

$\vec{B} = -\dfrac{\mu_0 I}{\pi^2 R}\hat{i}$

17. (4)

$\Delta Q = \dfrac{\Delta\phi}{R} = \dfrac{B\pi r^2}{R}$

$\Delta Q = 1.57 \times 10^{-4} C$

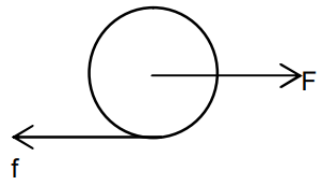
18. (4)

$(\Delta V_L + \Delta V_C)_{\max} = |\Delta V_L - \Delta V_C|$

$|2.6 - 7.9| = 4.8$ V.

19. (1)

$F = IBL$



$F - f = ma$

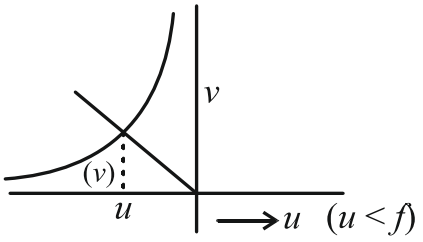
$\Rightarrow f = \dfrac{Bil - f}{2}$

$Fr = I\alpha$

$\Rightarrow f = \dfrac{Bil}{3} = 2N$

$a = r\alpha$

20. (1)



It is possible when object kept at centre of curvature.

$u = v$

$u = 2f, v = 2f$

21. **(2)**

$$\Rightarrow n(4I_0)=\frac{\int_{y=-\frac{\lambda D}{4d}}^{y=-\frac{\lambda D}{4d}} 4I_0\cos^2\frac{\pi dy}{\lambda D}dy}{\int_{-\frac{\lambda D}{4d}}^{\frac{\lambda D}{4d}} dy}$$

$$n=\frac{1}{2}+\frac{1}{\pi}$$

$$(2n-1)\pi=2$$

22. **(6)**

$\Rightarrow 0.529\times10^{-10}\, n^2 = 0.847\times10^{-9}$

$\Rightarrow n^2 = 16$

$n = 4$

$$\frac{n(n-1)}{2}=6$$

23. **(3)**

Change in momentum for each bullet fires is

$$=\frac{40}{1000}\times1200=48\,\text{N}$$

If a bullet fired exerts a force of 48 N on man's hand so ρ man can exert maximum force of 144 N, number of bullet that can be fired = 144/48 = 3 bullets.

24. **(4)**

$$\begin{bmatrix}\frac{hc}{\lambda} & =\phi+ev\\ \frac{hc}{2\lambda} & =\phi+\frac{eV}{3}\end{bmatrix}$$

$$\phi=\frac{hc}{4\lambda},\lambda_0=4\lambda$$

25. **(25)**

$$I=\frac{2}{80}=25\,\text{mA}$$

SECTION-II (CHEMISTRY)

26. **(4)**

$$\frac{M_1V_1}{n_1}(CO_3^{2-})=\frac{M_2V_2}{n_2}(H^+)$$

$$\Rightarrow\frac{M_1\times50}{1}=\frac{4\times49.35}{2}$$

$$\therefore M_1=1.974$$

$$M=\frac{x\times d\times10}{m_B}\Rightarrow1.974=\frac{x\times1.25\times10}{106}$$

$$\therefore x=16.7$$

27. **(2)**

$$\text{Time period}\propto\frac{n^3}{z^2}$$

$$\frac{T_1}{T_2}=\frac{n_1^3\times z_2^2}{n_2^3\times z_1^2};z_1=z_2 \text{ for same element.}$$

$$\text{Hence, }\frac{T_1}{T_2}=\frac{1^3}{3^3}=\frac{1}{27}$$

28. **(1)**

Due to poor shielding effect of 3d electron the size of Br^- ion nearly same as that of Cl^- ion.

29. **(1)**

Species	NO–	NO^+	CN^-	CN^+
Bond order	2	3	3	2

$$\text{Bondorder}\propto\frac{1}{\text{Bondlength}}$$

Bond length of NO^- > CN^+ due to greater number of antibonding electrons.

30. **(1)**

NO in iron complex has +1 oxidation number.

Thus, a + 5 × (0) + 1 + 1 × (–2) = 0

∴ a = +1

31. **(4)**

Free energy in an state function.

32. **(3)**

pH = 9.71, ∴ $[H^+] = 1.95\times10^{-10}$

$$Pr\,NH_2+HCl\rightleftharpoons Pr\,NH_3^++Cl^-$$

$$K_a=\frac{Pr[NH_2][H^+]}{[Pr\,NH_3^+]}$$

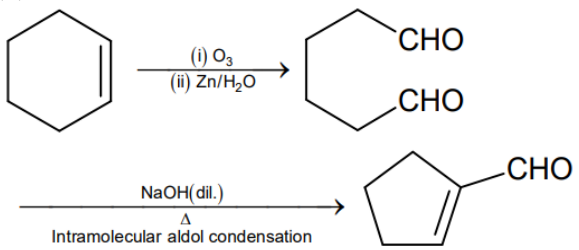
$$\frac{K_a}{[H^+]}=\frac{Pr[NH_2]}{[Pr\,NH_3^+]}$$

$$\Rightarrow\frac{[Pr\,NH_2]}{[Pr\,NH_3^+]}=0.1$$

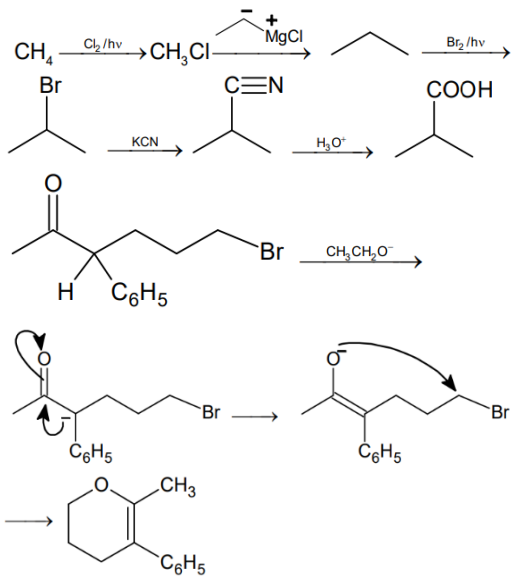
33. **(2)**

Because it has one d and one ℓ centre.

34. **(3)**



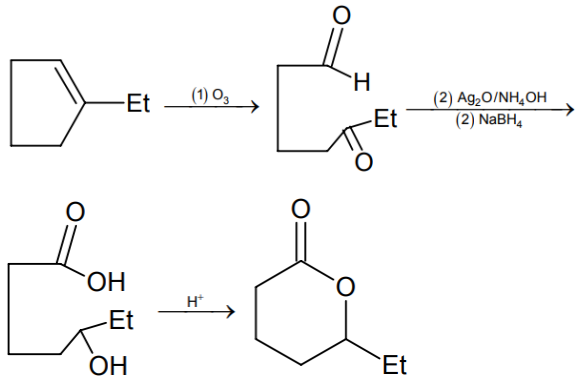
35. **(2)**



36. **(3)**

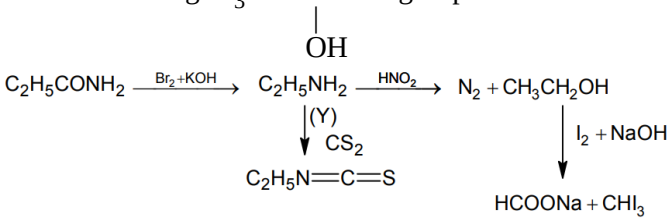
Acidity α-M effect

37. **(1)**

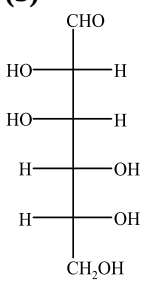


38. **(1)**

Reaction of X with Br_2 + KOH suggests that X is an amide. Evolution of N_2 and formation of alcohol suggests that Y is a 1° aliphatic amine. Iodoform test of Z suggests that it is an alcohol containing $H_3C-\underset{\underset{OH}{|}}{CH}-$ group.



39. **(3)**



C_2 epimer of D-Glucose

40. **(3)**

$$4Zn + 10HNO_3(dil) \rightarrow 4Zn(NO_3)_2 + 5H_2O + N_2O\uparrow$$

$$3Cu + 8HNO_3(dil) \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

$$Mn + 2HNO_3(dil) \rightarrow Mn(NO_3)_2 + H_2$$

$$Fe + 6HNO_3 \rightarrow Fe(NO_3)_3 + 3H_2O + 3NO_2$$

41 **(2)**

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow 2CrO_3 + 2KHSO_4 + 2H_2O$$

42 **(1)**

	G.I	**Stereoisomer**
$[Ma_3b_2c]$	3	3
$[Ma_3b_3]$	2	2
$[Ma_3bcd]$	4	5
$[Ma_4bc]$	2	2

43. **(3)**

$$P = P_A^\circ X_A + P_B^\circ = P_A^\circ(1 - X_B) + P_B^\circ X_B$$

$$\Rightarrow P = P_A^\circ - (P_A^\circ - P_B^\circ) \times B$$

Thus $P_A^\circ = 120$ Torr

$$P_A^\circ - P_B^\circ = 75$$

$$\Rightarrow P_B^\circ = 45 \text{ Torr}$$

Hence 3 is correct answer.

44. **(1)**

$$H_2O(\ell) + ClO_3^-(aq) \rightarrow ClO_4^-(aq) + 2H^+(aq) + 2e^-$$

$$2H^+(aq) + ClO_3^-(aq) + 2e^- \rightarrow ClO_2^-(aq) + H_2O(\ell)$$

$$2ClO_3^-(aq) \rightarrow ClO_2^-(aq) + ClO_4^-(aq)$$

$$E_{cell}^\circ = 0.33 - 0.36 = -0.03$$

$$E_{cell}^\circ = \frac{RT}{nF}\ln K \text{ (at equilibrium)}$$

$$-0.03 = \frac{0.059}{2}\log K$$

$\log K = -1$

$K = 0.1$

	$2ClO_3^-$	$\rightleftharpoons$	ClO_4^-	+	ClO_2^-
Initially	.1		0		0
Equilibrium	.1 – 2x		x		x

$$K = \frac{x^2}{(.1-2x)^2}$$

$$\frac{1}{10} = \frac{x^2}{(.1-2x)^2} \Rightarrow x = 0.019M$$

45. **(2)**

$$\ln\frac{k_1'}{k_1}=\frac{E_1}{R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$$

$$\text{en}\frac{k_2'}{k_2}=\frac{E_2}{R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$$

Eqn. (ii) - Eqn.(i)

$$\ln\frac{k_2'\times k_1}{k_2\times k_1'}=\frac{(E_2-E_1)}{R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$$

(For equimolar formation of B and C, $k_2'=k_1'$)

$$\ell n\left(\frac{k_1}{k_2}\right)=\left(\frac{8314}{8.314}\right)\left(\frac{T_2-300}{300\times T_2}\right)$$

$$\ell n2==\left(\frac{8314}{8.314}\right)\left(\frac{T_2-300}{300\times T_2}\right)$$

$T_2 = 379.77$

46. **(4)**

(S)-2,3- dimethylpentane, (R)-2,3-dimethylpentane, (S)-3-methylhexane and (R)-3-methylhexane.

47. **(3)**

$$KNO_3\rightarrow KNO_2+\frac{1}{2}O_2$$

$$2Pb(NO_3)_2\rightarrow 2PbO+4NO_2+O_2$$

$$NH_4NO_3\rightarrow N_2O+2H_2O$$

$$2Sr(NO_3)_2\rightarrow 2SrO+4NO_2+O_2$$

$$2Mg(NO_3)_2\rightarrow 2MgO+4NO_2+O_2$$

$$NaNO_3=NaNO_2+\frac{1}{2}O_2$$

$$Ca(NO_3)_2, Sr(NO_3)_2$$

$Pb(NO_3)_2$ do not respond to the test.

48. **(6)**

$$C_xH_{12}(g)+\left(x+\frac{12}{4}\right)O_2(g)\rightarrow$$

$$xCO_2(g)+6H_2O(g)$$

Initial moles of gaseous reactant taken

$= 1 + (x + 3) = x + 4$

Final moles of product formed = x + 6

Number of moles increased = 2

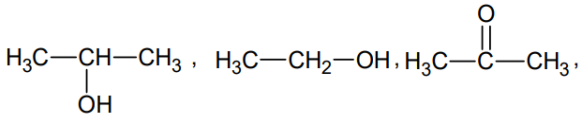
Pressure ∝ Number of moles.

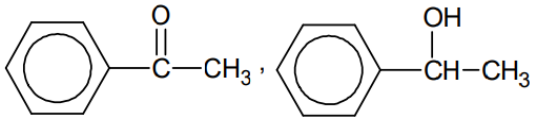
% of pressure increased

$$=\frac{2}{(x+4)}\times 100, 20=\frac{2}{(x+4)}\times 100, x=6.$$

49. **(5)**

The following compounds give positive iodoform test





50. **(2)**

Anode:

$$Cu(s)\rightarrow Cu^{+2}+2e^-$$

Cathode (1)

$$3e^-+4H^++NO_3^-\rightarrow NO+2H_2O$$

$$8H^-+2NO_3^-+3Cu(s)\rightarrow 3Cu^{+2}+2NO+4H_2O$$

$$Q=\frac{\left[Cu^{+2}\right]^3\times\left(p_{NO}\right)^2}{\left[NO_3^-\right]^2\left[H^+\right]^8}$$

$$E_{cel}^{\circ}=0.62$$

$$E_{cell}=1.3-\frac{0.059}{6}\log\frac{\left(Cu^{+2}\right)^3\left(p_{NO}\right)^2}{\left(NO_3^-\right)^2\times\left(H^+\right)^8}$$

Anode $Cu(s)\rightarrow Cu^{+2}+2e^-$

Cathode

$$e^-+2H^++NO_3^-\rightarrow NO_2+H_2O$$

$$Cu(s)+4H^++2NO_3^-\rightarrow 2NO_2+2H_2O+Cu^{+2}$$

$$E_{cell}^{\circ}=0.45$$

$$Q=\frac{\left(Cu^{+2}\right)\left(pNO_2\right)^2}{\left(NO_3^-\right)^2\left(H^+\right)^4}$$

$$E_{cell}=1.13-\frac{0.059}{2}\log\frac{\left(Cu^{+2}\right)\left(P_{NO}\right)^2}{\left(NO_3^-\right)^2\times\left(H^+\right)^4}$$

$$E_{cell_1}=E_{cell_2}$$

$$1.3-\frac{0.059}{6}\log\left(Q_1\right)=1.13-\frac{0.059}{2}\log\left(Q_2\right)$$

$$0.17=\frac{0.059}{6}\left\{\log\left(Q_1\right)-3\log\left(Q_2\right)\right\}$$

$$=\frac{0.059}{6}\left\{\log\frac{\left(Cu^{+2}\right)^3\times\left(p_{NO}\right)^2\times\left(NO_3^-\right)^6\left(H^+\right)^{12}}{\left(NO_3^-\right)^2\left(H^+\right)^8\times\left(Cu^{+2}\right)^3\times\left(p_{NO_2}\right)^6}\right\}$$

$$=\frac{0.059}{6}\left\{\log\frac{\left[NO_3^-\right]^4\left[H^+\right]^4}{\left(P_{NO_2}\right)^4}\right\}$$

$$0.17=\frac{0.059}{6}\times 8\log\left(HNO_3\right)$$

$$\log\left(HNO_3\right)=2.16$$

$$\left[HNO_3\right]=10^{2.16}=10^x$$

$x = 2.16 \Rightarrow \approx 2$

SECTION-III (MATHEMATICS)

51. (4)

We must have $f(a^2 + 2a + 5) > f(a + 11)$

$\Rightarrow \quad a^2 + 2a + 5 < a + 11$

$\Rightarrow a \in (-3, 2)$

$\therefore \quad a$ can take values $-2, -1, 0, 1$

52. (1)

Let $a - d = 3t$; then

$bc = (d + zt)(d + t) = d^2 + 3td + 2t^2$

Also $\frac{a}{d} = 1 + \frac{3t}{d}$,

so $hk = d^2\left[1 + \frac{3t}{d}\right] = d^2t + 3td < bc \Rightarrow bc > hk$

53. (2)

As $\sin x + \sin y \geq \cos\alpha \cos x \forall x \in R$

Let $x = -\frac{\pi}{2} \Rightarrow \sin y \geq 1 \Rightarrow \sin y = 1$

$\Rightarrow 1 + \sin x \geq \cos\alpha \cos x$

$\Rightarrow \cos\alpha \cos x - \sin x \leq 1$

$\Rightarrow \sqrt{\cos^2\alpha + 1} = 1 \Rightarrow \cos\alpha = 0$

$\Rightarrow \sin y + \cos\alpha = 1 + 0 = 1$

54. (2)

$I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \frac{x}{\sqrt{x}} dx = \frac{2}{3}$

$\Rightarrow I < \frac{2}{3}$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} < \int_0^1 x^{-1/2} dx = 2$

$\Rightarrow J < 2$

55. (1)

(A) Let E_i ; $i = 1, 2, 3, 4$ represents the events that the bag contains i white balls: Clearly $P(E_i) = \frac{i}{4}$. Let W be the event that the ball drawn is white then,

$$P(W) = \sum P(E_i) \cdot P\left(\frac{W}{E_i}\right) = \frac{1}{4}\left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4} + \frac{4}{4}\right) = \frac{5}{8}$$

Now $P\left(\frac{E_4}{W}\right) = \frac{P(E_4)P(W/E_4)}{P(W)}$

$= \frac{1/4}{5/8} = \frac{2}{5} = \frac{p}{15} \Rightarrow p = 6$

(B) $\frac{5x}{2-x} + \frac{5y}{2-y} + \frac{5z}{2-z}$

$= 5\left[\frac{x}{2-x} + \frac{y}{2-y} + \frac{z}{2-z}\right]$

$= 5\left[\frac{x-2+2}{2-x} + \frac{y-2+2}{2-y} + \frac{z-2+2}{2-z}\right]$

$= 5\left[-3 + 2\left[\frac{1}{2-x} + \frac{1}{2-y} + \frac{1}{2-z}\right]\right]$

Now apply A.M $\geq$ H.M on $2 - x$, $2 - y$ and $2 - z$

$\Rightarrow \frac{2-x+2-y+2-z}{3} \geq \frac{3}{\frac{1}{2-x} + \frac{1}{2-y} + \frac{1}{2-z}}$

$\Rightarrow \frac{1}{2-x} + \frac{1}{2-y} + \frac{1}{2-z} \geq \frac{9}{5}$

Hence $\frac{5x}{2-x} + \frac{5y}{2-y} + \frac{5z}{2-z} \geq 5\left[-3 + 2 \cdot \frac{9}{5}\right] = 3$

$\therefore$ Least value = 3

(C) $\sum_{K=1}^{n} 12 \cdot K^{12}C_K \cdot^{11} C_{K-1}$

$= 12^2 \sum_{K=1}^{12} \left({}^{11}C_{K-1}\right)^2 = 12^2 \cdot \frac{22!}{11!11!}$

$= 12 \cdot \frac{21 \cdot 19 \cdot 17 \ldots\ldots 3}{11!} - 2^{12} \cdot 6 \Rightarrow p = 6$

56. (1)

For non triangle solution $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$\Rightarrow \quad a + b + c = 0$

$\Rightarrow$ '1' is a root of $ax^2 + bx + c = 0$

57. (3)

Clearly we have

$\frac{xdy + ydx}{e^{xy}} = \frac{xdy - ydx}{x^2}$

$\Rightarrow \int d(xy)e^{-xy} = \int d\left(\frac{y}{x}\right) \Rightarrow \frac{y}{x} + e^{-xy} = c$

58. (2)

$xf'(x) + f(x) = 6f(x)f'(x)$

$f(x) = (6f(x) - x)f'(x)$

$I = \int \frac{2x(x - 6f(x)) + f(x)}{(6f(x) - x)\left(x^2 - f(x)\right)^2} dx$

$I = -\int \frac{2x}{\left(x^2 - f(x)\right)^2} dx + \int \frac{f(x)}{(6f(x) - x)\left(x^2 - f(x)\right)^2} dx$

$= \int \left(\frac{f'(x) - 2x}{\left(x^2 - f(x)\right)^2}\right) dx = \frac{1}{x^2 - f(x)} + c$

59. (1)

Let $g(x) = (\sin x)^{\ln x}$

$\Rightarrow g'(x) = \lim\limits_{h\to 0}\left(\dfrac{(\sin(x+h))^{\ln(x+h)} - (\sin x)^{\ln x}}{h}\right)$

$\Rightarrow f(x) = g'(x)$

$= g(x)\left(\dfrac{\ln(\sin x)}{x} + \cot x \ln x\right)$

$\Rightarrow f\left(\dfrac{\pi}{2}\right) = g'\left(\dfrac{\pi}{2}\right) = 0$

60. (2)

$n(x_1) = 3$, $x_1 = 2, 3, 4$

$4 \le x_2 < x_3 \le 8 \Rightarrow$ from 4, 5, 6, 7, 8

Selecting any 2 $\Rightarrow n(x_2, x_3) = {}^5C_2 = 10$

and $n(x_4) = 2$

$x_4 = 0, 5$

$\Rightarrow$ Number of such 4 digit numbers $3 \times 10 \times 2 = 60$

61. (4)

Let X denotes the number of toss required.

$P(X \ge 4)$

$= \left(\dfrac{2}{3}\right)^3 \cdot \dfrac{1}{3} + \left(\dfrac{2}{3}\right)^4 \cdot \dfrac{1}{3} + \left(\dfrac{2}{3}\right)^5 \cdot \dfrac{1}{3} + \ldots\infty$

$= \dfrac{\left(\dfrac{2}{3}\right)^3 \cdot \dfrac{1}{3}}{1 - \dfrac{2}{3}} = \dfrac{8}{27}$

62. (1)

$R = P'Q^8P = A^8$

$A^2 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} (\sqrt{3})^3 & -2(3) - 2\sqrt{3} - 2 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow$ If $[r_{ij}]_{2\times 2} = A^k$ then

$r_{11} = (\sqrt{3})^k \Rightarrow r_{11} = (\sqrt{3})^8 = 81$

63. (3)

$\lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \dfrac{1}{5^n} \cdot {}^nC_r \left(\sum\limits_{t=0}^{r-1} {}^rC_t \cdot 3^t\right)$

$= \lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \dfrac{1}{5^n} \cdot {}^nC_r \left(4^r - 3^r\right)$

$= \lim\limits_{n\to\infty} \dfrac{1}{5^n}\left(\sum\limits_{r=1}^{n} {}^nC_r 4^r - \sum\limits_{r=1}^{n} {}^nC_r 3^r\right)$

$= \lim\limits_{n\to\infty} \dfrac{1}{5^n}\left(5^n - 4^n\right) = 1$

64. (2)

$f\left(\dfrac{3\pi}{2}\right) = 2\pi$

$\Rightarrow f'\left(\dfrac{3\pi}{2}\right) \cdot g'(2\pi) = 1$

$\Rightarrow f'(x) = 25(2)(2x - 3\pi)^{24} + \dfrac{4}{3} - \sin x$

$\Rightarrow f'\left(\dfrac{3\pi}{2}\right) = \dfrac{7}{3} \Rightarrow g'(2\pi) = \dfrac{3}{7}$

65. (2)

$1 + 18\,(2) = 37$

66. (1)

Circle

67. (4)

(1, 1)

68. (3)

Given $x + y + z = 5 \rightarrow (1)$

and $x^2 + y^2 + z^2 = 9 \rightarrow (2)$

We have $yz = \dfrac{1}{2}\left[(y+z)^2 - \left(y^2 + z^2\right)\right]$

$= \dfrac{1}{2}\left[(5-x)^2 - \left(9 - x^2\right)\right]$ (using (1) & (2))

$= x^2 - 5x + 8$

$\Rightarrow$ y, z must be the roots of the equation

$u^2 - (5-x)u + \left(x^2 - 5x + 8\right) = 0 \rightarrow (3)$

As y, z are real, we must have discriminant of (3) ≥ 0

$\Rightarrow (5-x)^2 - 4\left(x^2 - 5x + 8\right) \ge 0$

$\Rightarrow 3x^2 - 10x + 7 \le 0$

$\Rightarrow (3x - 7)(x - 1) \le 0$

$\Rightarrow x \in \left[1, \dfrac{7}{3}\right]$

$\Rightarrow$ Number of integer values that x can take is 2

69. (3)

$2\sin^2 x + \sin^2 2x = 2$

$\Rightarrow -2\left(1 - \sin^2 x\right) + 4\sin^2 x \cos^2 x = 0$

$\Rightarrow \cos^2 x \cdot \cos 2x = 0 \rightarrow (1)$

$\sin 2x + \cos 2x = \tan x$

$\Rightarrow 2\sin x\cos^2 x+\cos 2x\cos x=\sin x$

$\Rightarrow \sin x\cos 2x+\cos 2x\cos x=0$

$\Rightarrow \cos 2x(\sin x+\cos x)=0\rightarrow(2)$

Common roots of (1) and (2) are given by cos2x = 0

$\Rightarrow x=\frac{(2n-1)\pi}{4}, n\in N$

In [0, 4π] we have $x=\frac{n\pi}{4}$, where n =1, 3, 5, …, 15 $\rightarrow$ (3)

Now, $-2\sin^2 x+\sin x\le 0$

$\Rightarrow 2\sin^2 x-\sin x\ge 0$

$\Rightarrow (\sin x)(2\sin x-1)\ge 0$

$\Rightarrow \sin x\in[-1,0]$ *or* $\frac{1}{2}\le\sin x\le 1\rightarrow(4)$

All the 8 values given by (3)

70. (3)

We have $w=z+\frac{1}{z}$

Take $z=e^{i\theta}(\because |z|=1)$.

We have $w=e^{i\theta}+e^{-i\theta}=2\cos\theta$

We know that w is real and $|w|\le 2$

71. (4)

$\frac{x-1}{x-4}>0$

$\Rightarrow x<1$ or $x>4$

$\Rightarrow$ Domain = $R-\{1,2,3,4\}$

72. (8)

Putting $X=x+1$ and $Z=z-1$

$\therefore f(X)+f(Z)=f(X+Z)$

$\therefore f'(X)=\lim_{h\to 0}\frac{f(X+h)-f(X)}{h}$

$=\lim_{h\to\infty}\frac{f(X)+f(h)-f(X)}{h}$

$=\lim_{h\to 0}\frac{f(h)-f(0)}{h}$

$=f'(0)=4\Rightarrow f'(X)=4$

$\Rightarrow f(X)=4X+c$

$\therefore f(x)=4x\Rightarrow f(2)=8$

73. (3)

Let $f(\theta)=\Pi(\tan\theta-\tan\alpha_i)-(\sum(\tan\theta-\tan\alpha_i))$

We have $f(\theta)_{\theta\to-\pi/2}\rightarrow-\infty$

Again $f(\alpha_1)>0, f(\alpha_3)<0$ and

$f(\theta)_{\theta\to\pi/2}\rightarrow\infty$

$\Rightarrow$ There will be 3 such values of 'θ'.

74. (5)

Let the equation be $K(x-\alpha)(x-\beta)=0$

$\therefore$ Sum of coefficient = $K(\alpha-1)(\beta-1)$ is prime

$\Rightarrow$ K is 1 and α – 1 is 1

$\Rightarrow$ One root is 2 and β – 1 is 2

Hence roots are 2 and 3.

75. (3)

Let $t=x^3$ in I_2

We have $I_2=\frac{1}{3}\int_0^1\frac{1}{e^t(2-t)}dt=\frac{1}{3}\int_0^1\frac{1}{e^{1-t}(1+t)}dt$

$\Rightarrow I_2=\frac{1}{3e}\int_0^1\frac{e^t}{(1+t)}dt=\frac{1}{3e}L_1\Rightarrow\frac{I_1}{eI_2}=3$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if